Assignment No. 11

1) Create a view that shows all of the customers who have the highest ratings.

```
KD3_rutuja_96226>CREATE VIEW v3_cust AS SELECT * FROM customers
WHERE rating = (select MAX(rating) FROM customers);
Query OK, 0 rows affected (0.094 sec)

KD3_rutuja_96226>SELECT * FROM v3_cust;
+------+----------+----------+--------+------+
| cnum | cname    | city     | rating | snum |
+------+----------+----------+--------+------+
| 2004 | Grass    | Berlin   |    300 | 1002 |
| 2008 | Cisneros | San Jose |    300 | 1007 |
+------+----------+----------+--------+------+
2 rows in set (0.126 sec)
```

2) Create a view that shows the number of salespeople in each city.

```
KD3_rutuja_96226>CREATE VIEW v0_sales AS SELECT city, COUNT(*) AS total_Sales
from salespeople GROUP BY city;
Query OK, 0 rows affected (0.322 sec)

KD3_rutuja_96226>select * from v0_sales;
+----------+-------------+
| city     | total_Sales |
+----------+-------------+
| London   |           2 |
| San Jose |           1 |
| Barcelona |           1 |
| New York |           1 |
+----------+-------------+
4 rows in set (0.036 sec)
```

3) Create a view that shows the average and total orders for each salesperson after his or her name. Assume all names are unique.

```
KD3_rutuja_96226>CREATE VIEW v_order AS SELECT s.sname, AVG(o.amt), count(o.snum)
FROM orders o INNER JOIN salespeople s ON o.snum = s.snum GROUP BY s.sname;
Query OK, 0 rows affected (0.142 sec)

KD3_rutuja_96226>select * from v_order;
+--------+------------+--------------+
| sname  | AVG(o.amt) | count(o.snum) |
+--------+------------+--------------+
| Rifkin |  558.425000 |            2 |
| Peel   | 5127.356667 |            3 |
| Motika | 1900.100000 |            1 |
| Serres | 1848.716667 |            3 |
| Axelrod | 1713.230000 |            1 |
+--------+------------+--------------+
5 rows in set (0.228 sec)
```

4) Create a view that shows each salesperson with multiple customers.

```
KD3_rutuja_96226>CREATE VIEW Salespeople_with_Multiple_Customers AS SELECT s.snum, s.sname
, COUNT(c.cnum) AS total_customers FROM Salespeople s JOIN Customers c ON s.snum = c.snum
GROUP BY s.snum, s.sname HAVING COUNT(c.cnum) > 1;
Query OK, 0 rows affected (1.432 sec)

KD3_rutuja_96226>select * from Salespeople_with_Multiple_Customers;
+------+--------+-----------------+
| snum | sname  | total_customers |
+------+--------+-----------------+
| 1001 | Peel   |               2 |
| 1002 | Serres |               2 |
+------+--------+-----------------+
2 rows in set (0.463 sec)
```

5) Which of these views are updateable (will allow DML operations)?

#1 Create View Dailyorders as Select Distinct cnum, snum, onum, odate from Orders;

```
KD3_rutuja_96226>select * from Dailyorders;
+------+------+------+------------+
| cnum | snum | onum | odate      |
+------+------+------+------------+
| 2008 | 1007 | 3001 | 1990-10-03 |
| 2001 | 1001 | 3003 | 1990-10-03 |
| 2007 | 1004 | 3002 | 1990-10-03 |
| 2003 | 1002 | 3005 | 1990-10-03 |
| 2008 | 1007 | 3006 | 1990-10-03 |
| 2002 | 1003 | 3009 | 1990-10-04 |
| 2004 | 1002 | 3007 | 1990-10-04 |
| 2006 | 1001 | 3008 | 1990-10-04 |
| 2004 | 1002 | 3010 | 1990-10-04 |
| 2006 | 1001 | 3011 | 1990-10-04 |
+------+------+------+------------+
10 rows in set (0.367 sec)

KD3_rutuja_96226>INSERT INTO Dailyorders VALUES(2010, 3012, '1990-10-05');
ERROR 1471 (HY000): The target table Dailyorders of the INSERT is not inser
table-into
```

#2 Create View Custotals as Select cname, Sum (amt) Sum_Amt from Orders, Customers where Orders.cnum = Customers.cnum Group by cname;

```
KD3_rutuja_96226>Create View Custotals1 as Select cname, SUM(amt) Sum_Amt from
 Orders, Customers where Orders.cnum = Customers.cnum Group by cname;
Query OK, 0 rows affected (0.058 sec)

KD3_rutuja_96226>select * from Custotals1;
+----------+----------+
| cname    | Sum_Amt  |
+----------+----------+
| Cisneros |  1116.85 |
| Hoffman  |   767.19 |
| Pereira  |  1900.10 |
| Liu      |  5160.45 |
| Giovanni |  1713.23 |
| Grass    |   385.70 |
| Clemens  | 14614.88 |
+----------+----------+
7 rows in set (0.022 sec)

KD3_rutuja_96226>DELETE from Custotals1 WHERE cname = 'Liu';
ERROR 1288 (HY000): The target table Custotals1 of the DELETE is not updatable
```

#3 Create view Thirdorders as Select * from Dailyorders where odate=’1990-10-03’;

```
KD3_rutuja_96226>Create view Thirdorders as Select * from Dailyorders where odate
='1990-10-03';
Query OK, 0 rows affected (0.116 sec)

KD3_rutuja_96226>SELECT * FROM Thirdorders;
+------+------+------+------------+
| cnum | snum | onum | odate      |
+------+------+------+------------+
| 2008 | 1007 | 3001 | 1990-10-03 |
| 2001 | 1001 | 3003 | 1990-10-03 |
| 2007 | 1004 | 3002 | 1990-10-03 |
| 2003 | 1002 | 3005 | 1990-10-03 |
| 2008 | 1007 | 3006 | 1990-10-03 |
+------+------+------+------------+
5 rows in set (0.093 sec)

KD3_rutuja_96226>update Thirdorders SET cnum = 2026 WHERE cnum = 2008;
ERROR 1288 (HY000): The target table Thirdorders of the UPDATE is not updatable
```

#4 Create view Nullcities as Select snum, sname, city from Salespeople where city is NULL OR sname BETWEEN 'A' and 'MZ';

```
KD3_rutuja_96226>Create view Nullcities as Select snum, sname, city from Salespeople
 where city is NULL OR sname BETWEEN 'A' and 'MZ';
Query OK, 0 rows affected (0.271 sec)

KD3_rutuja_96226>select * from Nullcities;
+------+---------+----------+
| snum | sname   | city     |
+------+---------+----------+
| 1004 | Motika  | London   |
| 1003 | Axelrod | New York |
+------+---------+----------+
2 rows in set (0.123 sec)

KD3_rutuja_96226>DELETE from Nullcities WHERE sname = 'Motika';
Query OK, 1 row affected (0.206 sec)

KD3_rutuja_96226>select * from Nullcities;
+------+---------+----------+
| snum | sname   | city     |
+------+---------+----------+
| 1003 | Axelrod | New York |
+------+---------+----------+
1 row in set (0.011 sec)
```

6) Create a view of the Salespeople table called Commissions. This view will include only the snum and comm fields. Through this view, someone could enter or change commissions, but only to values between .10 and .20.

```
KD3_rutuja_96226>CREATE VIEW Commissions2 AS SELECT snum, comm from salespeople
    -> WHERE comm BETWEEN .10 AND .20 WITH CHECK OPTION;
Query OK, 0 rows affected (0.209 sec)

KD3_rutuja_96226>select * from Commissions2;
+------+------+
| snum | comm |
+------+------+
| 1001 | 0.12 |
| 1002 | 0.13 |
| 1007 | 0.15 |
| 1003 | 0.10 |
+------+------+
4 rows in set (0.130 sec)

KD3_rutuja_96226>UPDATE Commissions1 SET comm = .30
    -> WHERE snum = 1003;
ERROR 1369 (HY000): CHECK OPTION failed 'sales.commissions1'
KD3_rutuja_96226>UPDATE Commissions1 SET comm = .11
    -> WHERE snum = 1004;
Query OK, 0 rows affected (0.028 sec)
Rows matched: 0  Changed: 0  Warnings: 0
```